Kali Fang

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Mechanical Engineering senior specializing in hardware–software system integration, control systems, and electromechanical design. Experienced across the full development lifecycle, including sensor integration, signal processing, motor control, and UI-design. Seeking full-time engineering positions.

EDUCATION

Worcester Polytechnic Institute | *Worcester, MA* — Aug 2022 - May 2026

Bachelor of Science in Mechanical Engineering, Minor in Robotics Engineering

EXPERIENCE

Saint-Gobain | Engineering Intern - *Northborough, MA* — Jul 2025 - Dec 2025

- Designed and developed Python-based application to automate test results and data acquisition system to visualize data and perform statistical analysis (ANOVA, t-tests, regression), reducing analysis time by 90%
- Developed test protocols and conducted materials characterization using multiple instrumentation systems including SurPASS zeta potential analyzer, NIR spectroscopy, tensile/tear testers, and air permeability systems
- Collaborated with cross-functional teams (chemists, process engineers, manufacturing) to provide data analysis support, conducting statistical evaluations to guide product development decisions
- Authored technical documentation (SOPs, NKC reports) to report and standardize future testing and streamline data collection workflows and ensure repeatability and reproducibility across all users

WPI Chemistry Department | Chemical Stockroom Technician - *Worcester, MA* — Oct 2022 - May 2024

- Organized 200+ chemical inventory in Excel, ensuring safety compliance and operational efficiency
- Organized proper storage of temperature-sensitive and flammable substances, ensuring OSHA safety standards

PROJECTS

Autonomous SLAM Navigation System - *Worcester, MA* — Mar 2026 - May 2026

- Designed modular ROS2 architecture with custom nodes for frontier detection, path planning, and motion control
- Deployed GMapping-based SLAM pipeline integrating LiDAR data streams to construct and maintain real-time 2D occupancy grids in dynamic unknown environments
- Developed custom A* path planning algorithm with hybrid heuristics incorporating Euclidean distance, distance-to-wall penalties, and angle optimization for safe, efficient navigation
- Utilized OpenMV computer vision to filter and detect obstacles, augmenting sensor fusion inputs for real-time SLAM-based navigation

Automated and Accurate Blood Pressure Monitor - *Worcester, MA* — Aug 2025 - May 2026

- Integrated ESP32-based hardware subsystems via full-stack assembly and sensor calibration, using a rolling mean averaging for real-time signal processing under high-noise conditions following ISO 81060 and IEC 80601-2
- Designed a custom CAD enclosure to consolidate soldered components into a compact, mechanically stable assembly compliant with ISO standards
- Achieved 61.5% of readings within 5 mmHg, outperforming an FDA-approved device by 17%

Robotic Arm Control System (4-DOF) - *Worcester, MA* — Aug 2024 - Oct 2024

- Worked with a team of 3 to develop real-time motion control system in Linux using MATLAB/C++ for 4-DOF robotic manipulator, achieving 90% task accuracy through closed-loop control and camera-based vision feedback
- Developed custom software libraries and optimized control algorithms for kinematics calculations, trajectory planning, and motor control

Autonomous Robotic Maze Navigator - *Worcester, MA* — Mar 2024 - May 2024

- Designed and programmed real-time control system using an ESP32, integrating WiFi communication, IMU sensors, and vision recognition system for multi-robot coordination
- Implemented H-bridge motor driver circuits interfacing digital microcontroller control signals with analog motor actuation for closed-loop motion control across 4 robots
- Implemented PID control algorithms for ultrasonic distance sensors and motor control, improving navigation accuracy by 34% through sensor fusion and real-time feedback

SKILLS

Programming Languages: C/C++, C#, Python, MATLAB, Simulink, JavaScript, SQL

Embedded Systems: Microcontrollers, Arduino (C/C++), ESP32, Raspberry Pi, PlatformIO, Linux

Control Systems: PID Control, Motion Control, Sensor Fusion, Multi-threading, Real-Time Data Processing

Software: Linux, SolidWorks, CAD, FEA, Git/GitHub, Visual Studio, Streamlit
Hardware & Testing: Oscilloscope, Logic Analyzer, Stepper/Servo Motor Control, Serial Communication (UART, I2C, SPI), Soldering